

Jeric Lew Jieyi

Research Assistant (Robotics)

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Education

National University of Singapore (NUS), Singapore

Master of Engineering (Mechanical Engineering) | GPA: 5.0/5.0 Aug 2025 – May 2027

Bachelor of Engineering (Mechanical Engineering) | GPA: 4.9/5.0 Aug 2021 – May 2025

Georgia Institute of Technology, Atlanta, GA

Undergraduate Study Abroad (Non-Degree) | GPA: 4.0/4.0 Jan 2024 – May 2024

Research Experience

MARMot Lab, National University of Singapore (NUS), Singapore

Research Assistant & Master's Thesis with [Guillaume Sartoretti](#) Aug 2025 – Present

- Research on unsupervised reinforcement learning (RL) with World Models for single/multi-agents
- Mentoring thesis students on diffusion models and RL for path planning

Research Intern & Bachelor's Thesis with [Guillaume Sartoretti](#) & [Cao Yuhong](#) Aug 2023 – Jul 2025

- Research on diffusion models and RL for path planning in single/multi-agent robot systems
- Developed and tested novel deep-learning methods in high fidelity simulations and real robots

AirLab, Carnegie Mellon University (CMU), Pittsburgh, PA

Robotics Research Intern with [Sebastian Scherer](#) & [Wenshan Wang](#) Jun 2024 – Aug 2024

- Research on [off-road navigation](#) as a part of CMU's [Robotics Institute Summer Scholars](#) program
- Distilled vision foundation models (DINOv2) for faster inference and higher feature resolution [2]
- Integrated pipeline into a large, complex, multi-language off-road navigation stack with ROS [3]

Selected Projects

Unsupervised Skill Discovery with World Models Work In Progress

Master's thesis aimed at unsupervised learning of skills with downstream composition via world models

- Implemented highly parallelised reinforcement learning environments with JAX
- Preliminary results with world model composition of discovered skills for a multi-agent team

Diffusion Models for Informative Path Planning (IPP) github.com/marmotlab/DARE & [AID](#)

Bachelor's thesis on leveraging long horizon diffusion trajectories for single and multi agent IPP

- Achieved strong performance with behaviour cloning on privileged expert data for single agent IPP [4]
- Enabled multi-agent coordination by sharing long horizon trajectories and RL finetuning (DPPO) [5]
- Enabled test-time multi-agent coordination with diffusion models trained with diverse behaviour [6]

PPO for Block Pushing Task with Robot Arm github.com/JericLew/Push_MuJoCo

Graduate class on deep RL for robotics

- Designed a custom MuJoCo simulation environment for block pushing task
- Teacher-Student distillation from RL trained privileged state-based policy to vision-based policy
- Attained 70% success rate from with student policy trained with behaviour cloning without expert demonstrations

Skills

Programming Languages: Python, C/C++ , MATLAB, Java, ARM-7 Assembly

Frameworks/Tools: PyTorch, JAX, ROS1/2, MuJoCo, CUDA, OpenCV, Arduino

Hardware: SOLIDWORKS, 3D Printing, Machining, Microcontroller, Blender

Concepts: Reinforcement Learning (PPO, SAC, etc.), Imitation Learning (Diffusion, Flow Matching), Computer Vision (UNet, ViT, DINOv2), Planning (A*, D*, RRT*, etc.)

Selected Publications

[6] Planner-Conditioned Diffusion for Coordinated Multi-Agent Exploration

Marcus Yu Siong Teo*, **Jeric Lew***, Tanishq Duhan, Guillaume Sartoretti

International Symposium on Distributed Autonomous Robotic Systems (DARS), 2026 | To Be Presented

[5] AID: Agent Intent from Diffusion for Multi-Agent Informative Path Planning

Jeric Lew, Yuhong Cao, Derek Ming Siang Tan, Guillaume Sartoretti

International Conference on Swarm Intelligence (ANTS), 2026 | [Code](#)

[4] DARE: Diffusion Policy for Autonomous Robot Exploration

Yuhong Cao*, **Jeric Lew***, Jingsong Liang, Jin Cheng, Guillaume Sartoretti

IEEE International Conference on Robotics and Automation (ICRA), 2025 | [Code](#) | [Video](#)

[3] SALON: Self-supervised Adaptive Learning for Off-road Navigation

Matthew Sivaprakasam, Samuel Triest, Cherie Ho, Shubhra Aich, **Jeric Lew**, Isaiah Adu, Wenshan Wang, Sebastian Scherer

IEEE International Conference on Robotics and Automation (ICRA), 2025 | [Project](#) | [Video](#)

[2] SHRED: Swift High-Resolution features via Efficient Distillation

Jeric Lew, Matthew Sivaprakasam, Samuel Triest, Wenshan Wang, Sebastian Scherer

RISS Working Papers Journal, 2024 | [Poster](#) | [Video](#)

[1] A novel application for real-time arrhythmia detection using YOLOv8

Guang Jun Nicholas Ang, Aritejh Kr Goil, Henryk Chan, **Jieyi Jeric Lew**, Xin Chun Lee, Raihan Bin Ahmad Mustaffa, Timotius Jason, Ze Ting Woon, Bingquan Shen

arXiv, 2024

Teaching

Teaching Assistant, [ME1102 Engineering Principles and Practice I](#)

Fall 2024

Teaching Assistant, [EG1311 Design and Make](#)

Summer 2022 - Fall 2023

Teaching Assistant, [CS1010E Programming Methodology](#)

Fall 2022

Awards and Honors

Innovation & Research Award

Spring **2025**

Best Student in Microprocessor Applications

Spring **2023**

Dean's List

Fall **2022, 2023, 2024**

NUS Merit Scholarship

Fall 2021 - Spring 2025